Appendix B - South-west Corner (Western arm) Marine Park Pressures Report

South-west Marine Parks Network

Table of contents

Pressures 1

Pressures

Understanding of Pressures

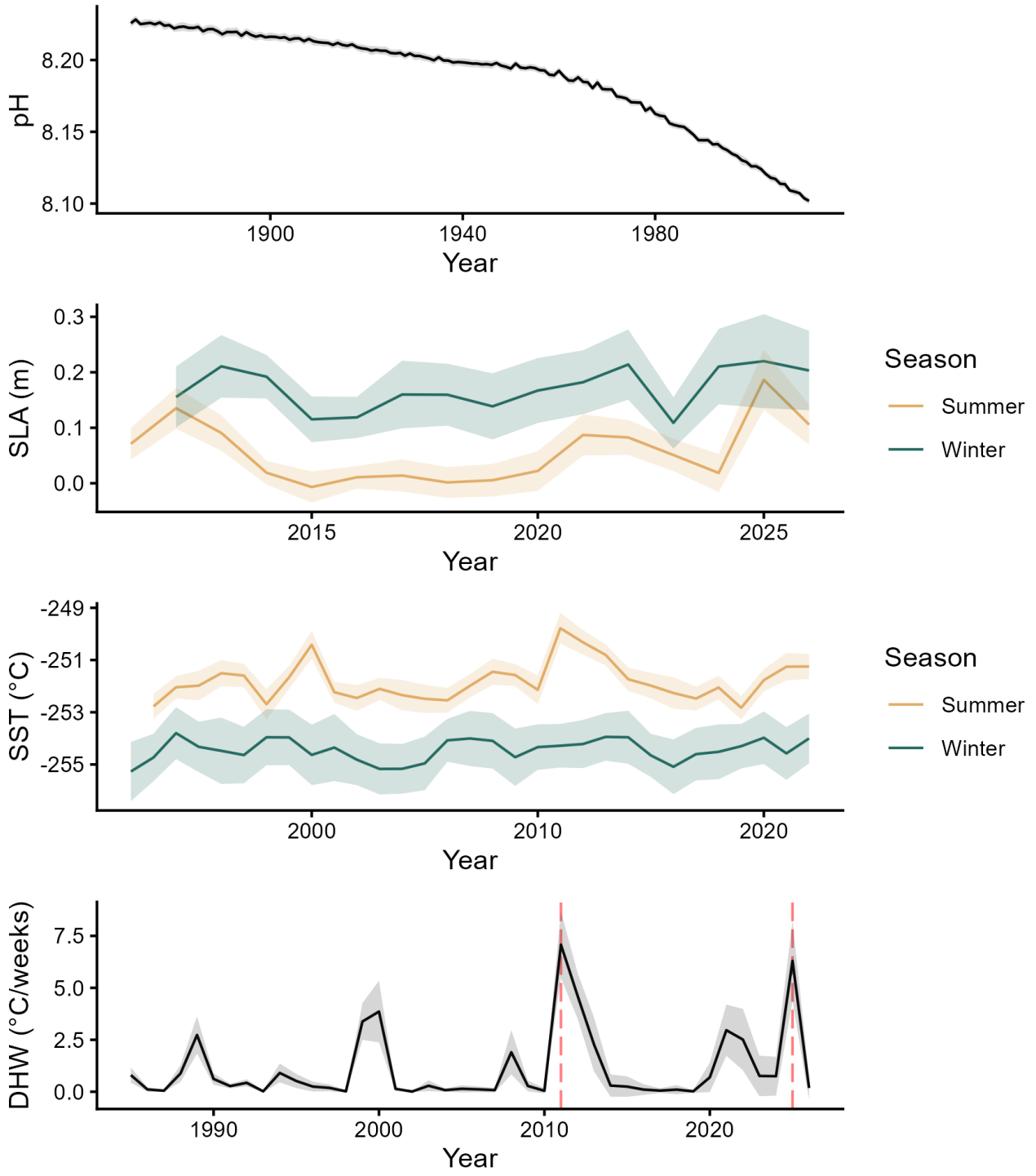


Figure B.1: Seasonal sea level anomaly, seasonal sea surface temperature and Degree Heating Weeks at the scale of the SWCMP (Western arm). Solid lines represent mean values for sea level anomaly (SLA), sea surface temperature (SST) and Degree Heating Weeks (DHW). Confidence bounds represent standard errors. Red dashed vertical lines represent maximum DHW values that are plotted spatially in Figure B.4.

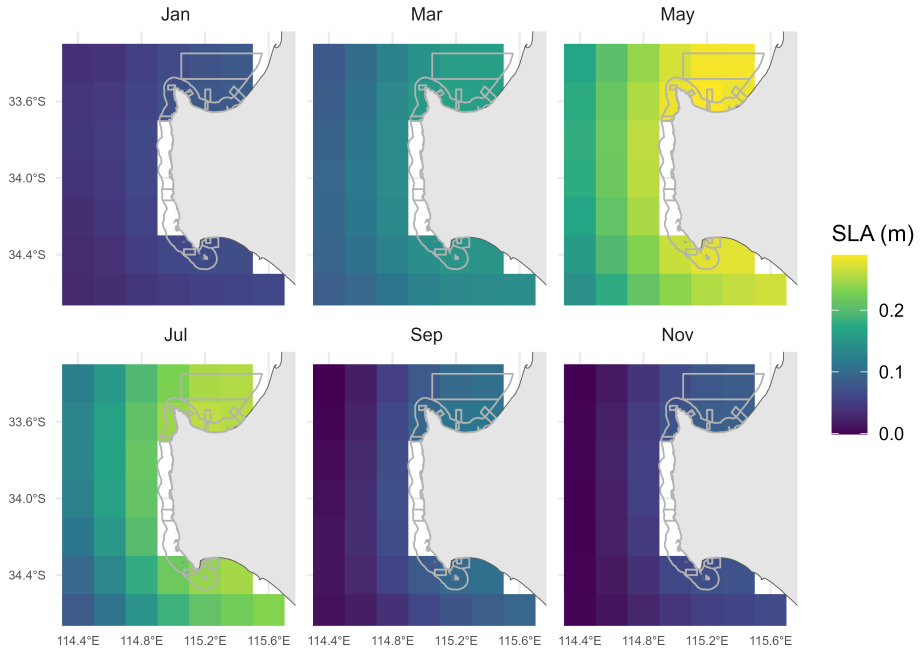


Figure B.2: Sea level anomaly, as an index of upwelling, at the scale of the SWCMP (Western arm). Heatmap values represent high (yellow) and low (blue) bimonthy sea level anomaly (SLA). Grey polygons represent State spatial management and Australian Marine Park boundaries.

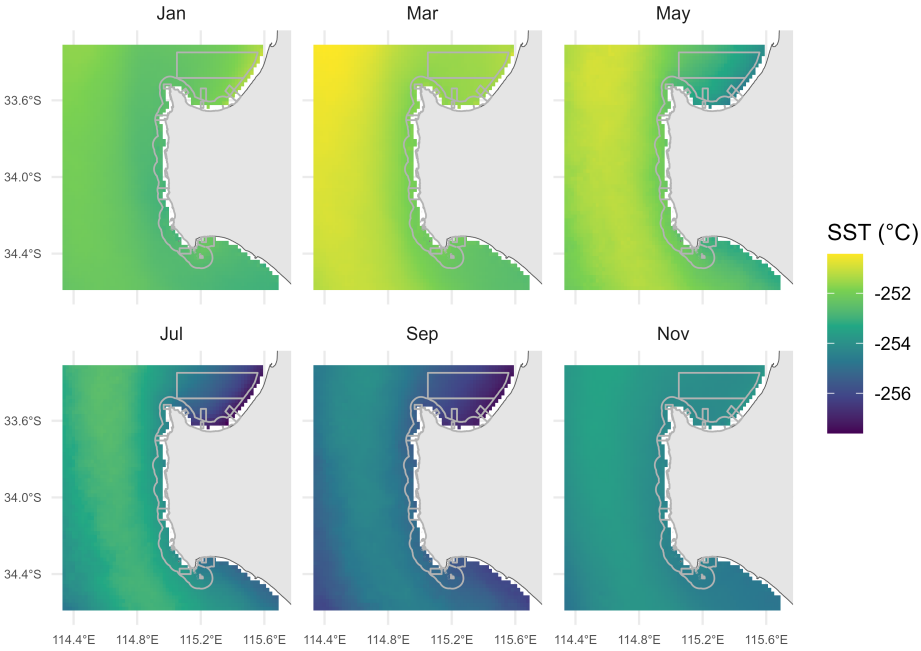


Figure B.3: Sea surface temperature at the scale of the SWCMP (Western arm). Heatmap values represent high (yellow) and low (blue) bimonthy sea surface temperature (SST). Grey polygons represent State spatial management and Australian Marine Park boundaries.

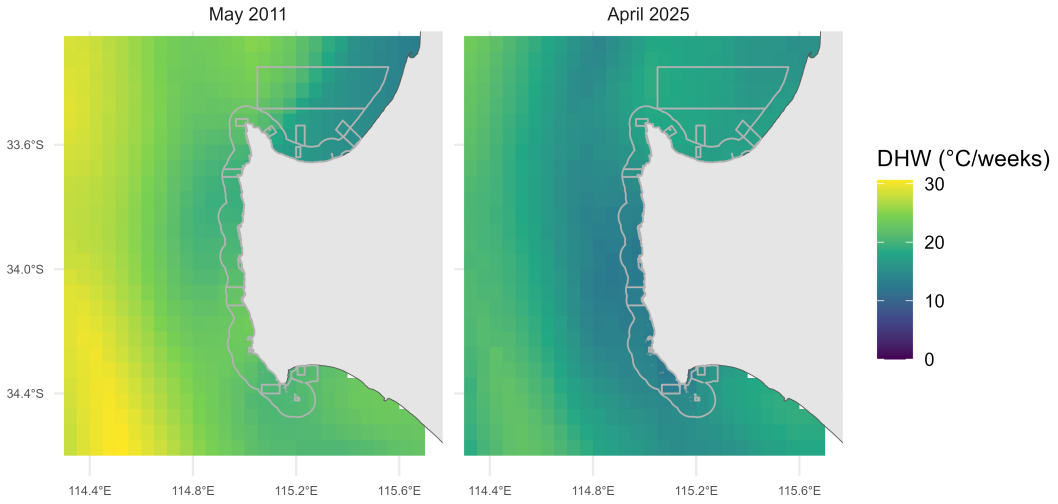


Figure B.4: Maximum Degree Heating Weeks at the scale of the SWCMP (Western arm). Heatmap values representing high (yellow) and low (blue) monthly Degree Heating Weeks (DHW) for the two most recent marine heatwave periods on record. Grey polygons represent State spatial management and Australian Marine Park boundaries.

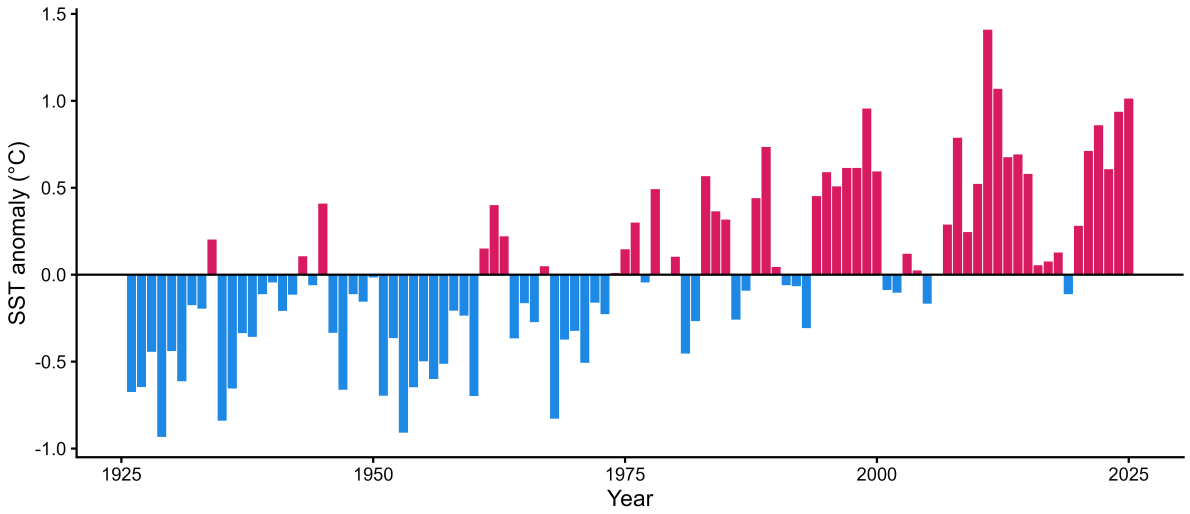


Figure B.5: Annual sea surface temperature anomaly at the scale of the SWCMP (Western arm) over the last 100 years, relative to the 1961-1990 average. Red bars represent years warmer than average and blue bars years cooler than average. Data from NOAA ERSST v6, a 2° monthly reconstruction from ship, buoy and Argo float observations, averaged over the 2 ERSST grid cells (113-117°E, 33-35°S) overlapping the park; values before 1940 are less certain due to sparse observations.